

A Fully Analog Computing-in-Memory Macro With INT8-MAC Operations for Edge-AI Device

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Abstract—Sensors capture large volumes of data containing both critical and redundant information for neural network (NN) inference. However, limited bandwidth and energy efficiency at the sensor-side analog-to-digital conversion (ADC) and data transmission stages significantly increase latency and power consumption in edge devices. Conventional data compression can reduce data volume but cannot distinguish which information is important for the NNs, especially in early layers, where precision is highly sensitive. Such distortions propagate and degrade the overall inference accuracy. To address these challenges, this paper proposes a fully analog computing-in-memory (FACIM) macro that implements and experimentally demonstrates analog-domain feature extraction and inter-layer analog signal propagation mechanisms for NNs. Unlike conventional CIM architectures that rely on inter-layer ADCs, the FACIM concept performs feature extraction directly in the analog domain, thereby reducing early-stage quantization effects and unnecessary data movement. This approach provides a potential pathway toward improved energy efficiency in edge-AI systems. The FACIM macro was fabricated using TSMC 0.18- μm CMOS 1P6M standard technology. It achieves an energy efficiency of 1,880 TOPS/W for INT8 multiply-and-accumulate (MAC) operations and 117 TOPS/W at the system level. The macro demonstrates an area efficiency of 97.76 GOPS/ mm^2 [normalized macro area (Jhang, 2021)].

Index Terms—Fully analog CIM, ADC-free CIM, near sensor computing, AI accelerators, edge AI, analog neural network, SRAM.

I. INTRODUCTION

IN ARTIFICIAL intelligence of things (AIoT) sensing nodes, sensors acquire large volumes of environmental data, much of which is redundant or irrelevant, leading to transmission congestion. With limited data communication bandwidth, such congestion reduces frame rates and energy efficiency, a phenomenon analogous to the Von Neumann bottleneck. To alleviate this issue, edge artificial intelligence (Edge-AI) techniques perform certain data processing tasks such as compression and feature extraction directly at the edge devices, thereby reducing transmission volume, improving real-time responsiveness, and enhancing data privacy [2]. However, conventional computing-in-memory (CIM) architectures often rely on analog-to-digital converters (ADCs)

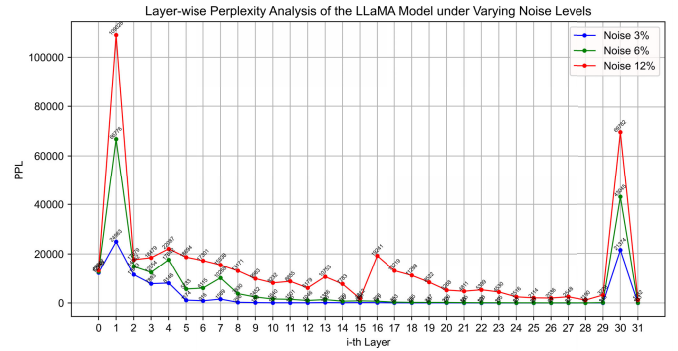


Fig. 1. Layer-wise noise sensitivity analysis.

for inter-layer data transfer, which introduces quantization loss and significantly increases data movement power consumption. The proposed Fully Analog Computing-in-Memory (FACIM) architecture directly addresses these challenges by avoiding quantization loss and reducing data movement power consumption.

Although digital computing-in-memory (DCIM) architectures have been applied in sensors for real-time data encoding [3], [4]. This approach effectively compresses transmitted data, accelerates transmission, and reduces power consumption. However, most existing encoding methods treat all data equally without distinguishing between critical and non-critical features for neural network (NN) inference. As a result, important features may be compressed or discarded, which can negatively affect inference accuracy. Furthermore, digital processing necessitates prior data quantization. This quantization process introduces noise that may obscure subtle yet crucial features, particularly in the early layers of the NN where high precision is essential. This ultimately degrades the overall performance and accuracy of the neural network.

It's crucial to emphasize that the early layers of a neural network play a critical role in determining overall inference accuracy. These layers are primarily responsible for extracting low-level features directly from the raw data. If these initial features are distorted by noise, quantization errors, or data loss, all subsequent layers will process flawed representations. This leads to a compounding of errors across the network layers, ultimately resulting in a significant degradation of the model's final prediction accuracy.

As illustrated in Fig. 1, an analysis of MAC precision in the Llama2 model using CIM with an 8-bit ADC was conducted.

Received 19 August 2025; revised 19 December 2025; accepted 19 January 2026. Date of publication 22 January 2026; date of current version 15 June 2026. This article was recommended by Guest Editor D. Fan. (*Corresponding author: Wei-Cheng Huang.*)

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Digital Object Identifier 10.1109/JETCAS.2026.3657011

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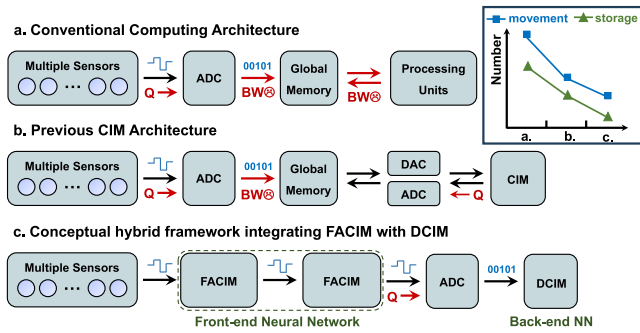


Fig. 2. Various architectures of AI accelerators.

The related study investigated the impact of ± 1 LSB error rates of 3%, 6%, and 12% during MAC operations across different layers. The results reveal that the first few layers and the last two layers of Llama2 exhibit higher perplexity, indicating their critical importance to the model's inference performance. Similar observations have been reported in other neural network models [5], [6], [7], [8], [9]. Consequently, to preserve inference accuracy, these initial layers are generally maintained at higher precision and subjected to lower quantization errors.

To address these challenges, this paper proposes a Fully Analog Computing-In-Memory (FACIM) macro for Edge-AI applications, supporting INT8 MAC operations with analog I/O. FACIM is based on an analog neural network architecture combined with an analog transmission circuit to reduce quantization noise and perform initial feature extraction directly in the analog domain. This approach effectively lowers data transmission volume and enhances overall system efficiency. Simulation results demonstrate that the proposed analog transmission circuit achieves an effective MAC precision equivalent to 10 bits (input signal/input referred noise $> 2^{10+1}$), even under noise interference and PVT variations. Compared to conventional architectures, FACIM substantially reduces the movement of redundant data and greatly minimizes dependence on ADCs and registers.

As shown in Fig. 2, this paper presents a conceptual hybrid framework that integrates FACIM with a conventional CIM architecture to combine the advantages of both. FACIM performs analog feature extraction directly at the edge device, significantly reducing data dimensionality and improving energy efficiency, whereas the conventional CIM architecture provides greater flexibility in neural network implementation. By leveraging these complementary strengths, the hybrid solution has the potential to deliver both high performance and adaptability for edge-AI applications. In this paper, only the FACIM architecture is implemented, with a focus on validating the feasibility of analog feature extraction and inter-layer analog signal transmission.

The remainder of this paper is organized as follows. Section II provides an overview of the multi-bit CIM macro. Section III introduces the structure of the proposed FACIM macro. Section IV discusses training-hardware consistency in neural network systems. Section V presents the performance and measurement results. Finally, Section VI concludes this paper.

II. OVERVIEW OF MULTI-BIT CIM MACRO

The increasing complexity of datasets has driven the development of multi-bit CIM architectures to enhance inference accuracy [10], [11], [12], [13], [14]. As illustrated in Fig. 3(a), conventional multi-bit CIM designs perform MAC operations in two steps: 1) executing partial MAC operations on low-bit weights and input features; 2) quantizing these partial results using multiple parallel ADCs and combining them into the complete multi-bit MAC value via digital shifters and adders (DSAs). The FACIM architecture proposed in this paper also employs a two-stage MAC process. However, unlike conventional designs, FACIM preserves and transmits features entirely in the analog domain. This section will first review conventional multi-bit CIM architectures before presenting the FACIM solution and its advantages.

A. Partial MAC Operations

Analog CIM designs utilize various methods for partial MAC operations, including current discharge [10], [12], charge sharing and coupling [11], [14], and timing delay [13].

Current-domain CIM architectures typically integrate MAC operations directly into existing ReRAM or SRAM circuits, offering excellent area efficiency. However, the current discharge mechanism is relatively complex and faces limitations in linearity, making it suitable only for low-accumulation MAC applications.

For instance, in ReRAM-based CIM systems [10], the discharge current varies with the voltage across the resistive cell, resulting in an exponential discharge curve. This leads to significant nonlinearity in the MAC results. Consequently, such architectures are restricted to low-accumulation operations and require compensation using nonlinear calibration tables.

In contrast, SRAM-based current-domain CIM [12] utilizes transistors to provide a relatively stable discharge current. Nevertheless, the drain voltage fluctuates depending on the number of activated MAC cells, causing variations in the output MAC voltage (VMAC) and introducing nonlinearity. To maintain acceptable linearity, VMAC must be constrained within a specific range, which in turn limits the available signal margin. Overall, while these current-domain approaches excel in area efficiency, their constrained signal margin reduces the effectiveness of analog CIM in high-accumulation scenarios.

To improve the signal margin, charge-domain CIM adopts charge redistribution techniques for MAC operations, which can be implemented using either charge sharing or charge coupling. In charge-sharing architectures, charge is transferred through switches. As the number of switches increases, linearity degrades significantly, primarily due to the parasitic capacitance of MOS switches. This capacitance is highly sensitive to PVT variations and introduces charge injection effects. As demonstrated in [11], employing a large number of switches for charge sharing limits linearity and makes it difficult to maintain MAC precision, ultimately preventing charge-domain CIM architectures from realizing their full potential.

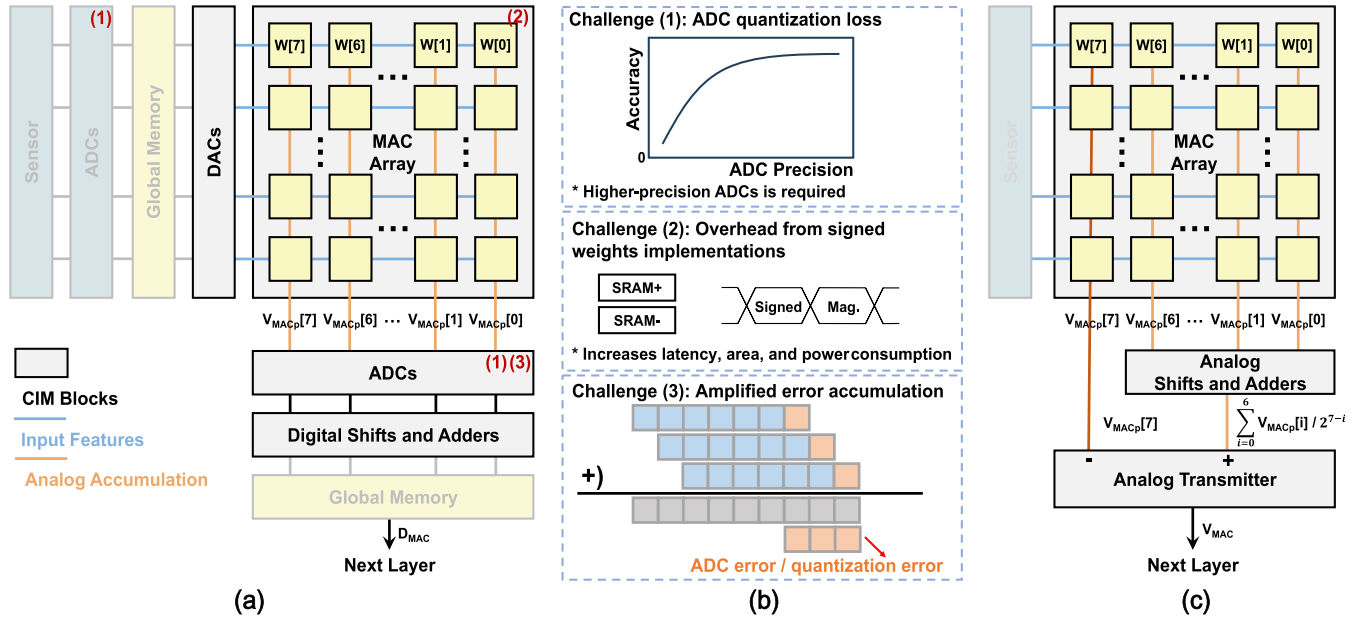


Fig. 3. (a) Typical multi-bit CIM structures. (b) Challenges in multi-bit CIM structures. (c) Proposed INT8 FACIM structures.

To further enhance signal margin, two alternative CIM architectures have been proposed: charge-coupling CIM [14] and time-domain CIM [13]. However, these approaches typically come at the cost of reduced area efficiency.

In charge-coupling CIM [14], analog signals are coupled using metal-oxide-metal (MOM) capacitors when the SRAM-stored weight is “1”. These capacitors exhibit strong immunity to PVT variations and effectively avoid nonlinearities induced by switching elements. Consequently, charge coupling achieves superior linearity and enhances accumulation capability.

On the other hand, time-domain CIM maps the MAC operation to time delays, allowing for an extended signal range. However, its accuracy is strongly dependent on process technology and operating speed. The delay cells employed in this approach are highly sensitive to PVT variations and device mismatches, often introducing timing errors. To mitigate these issues, larger and slower delay cells are typically employed. For instance, in [13], each delay cell occupies an area equivalent to 0.6 times that of 32 standard 6T SRAM cells, significantly compromising overall area efficiency.

Considering the trade-off between area efficiency and computational linearity, this paper adopts a 9T1C SRAM-based MAC cell architecture similar to that in [15], and [16]. This design performs MAC operations between analog input features and 1-bit weights using a charge-coupling approach. The design details and area efficiency considerations of the 9T1C SRAM-based MAC cell will be discussed in the next section, while the impact of MOM capacitor mismatch in advanced technology nodes will be addressed in Section IV.

B. Multi-Bit MAC Operations

As depicted in Fig. 3(a), most multi-bit CIM designs [10], [11], [12], [13], [14] employ multiple parallel ADCs

to quantize partial V_{MAC}, which are then combined using a digital shifter-adder (DSA) to produce the full multi-bit MAC result. However, mismatches between parallel ADCs are inevitable. Non-idealities such as offset errors, gain errors, quantization noise, and other disturbances become amplified and accumulated during the DSA process. Consequently, the overall precision of the CIM system is fundamentally constrained, particularly by the ADC responsible for handling the most significant bits (MSB), leading to degraded neural network inference accuracy, as illustrated in Fig. 3(b). Moreover, achieving higher ADC resolution introduces significant power and latency overhead, which contradicts the low-power and high-energy-efficiency design objectives of CIM systems.

Furthermore, many previous CIM architectures [10], [11], [12], [13], [14] do not natively support signed operations. This necessitates the separation of positive and negative weights during neural network training, which reduces model flexibility and can compromise inference accuracy, as shown in Fig. 3(b). On the hardware side, signed operations are often handled by processing positive and negative components of features and weights in separate cycles or through spatial partitioning. However, this approach increases latency, area overhead, and power consumption, ultimately limiting system performance.

To overcome these limitations, this paper proposes a FACIM architecture, which performs feature extraction and MAC operations entirely in the analog domain. As a result, the conventional DSA used in prior CIM architectures is not applicable. To support these analog MAC operations, this paper introduces an Analog Shifter-Adder (ASA). The ASA partitions the MAC capacitor array based on weight ratios. Leveraging charge-sharing techniques, the ASA enables efficient multi-bit analog MAC operation with minimal area and

power consumption, achieving high performance within a short time.

The proposed capacitance ratio-based accumulation approach mitigates noise interference and prevents the amplification of quantization errors during the accumulation process, thereby enhancing the accuracy of the output features. Furthermore, by employing a 2's complement format for weight storage, the system can capture both positive and negative partial MAC results within the MAC capacitor array, as shown in Fig. 3(c). When combined with the analog feature transmission circuit in FACIM, this mechanism enables seamless signed operations. The circuit-level implementation details will be discussed in the next section.

III. STRUCTURE OF THE PROPOSED FULLY ANALOG CIM MACRO

A. Overall Structure of Proposed FACIM Macro

Fig. 3(c) presents the structure of a single neuron in the proposed FACIM architecture. The FACIM operates in two modes: memory mode and CIM mode. In memory mode, the stored weights are accessed using a peripheral circuit and written into the MAC cell in 2's complement data format.

As shown in Fig. 4(a), the proposed FACIM architecture operates in the CIM mode. Within this configuration, analog features are initially processed in the voltage domain before being converted for MAC operations executed in the charge domain by the CIM array and the Analog Shifter-Adder (ASA). The resulting output voltage is then transmitted to activate the subsequent neural network layer, thereby establishing the end-to-end data flow of the FACIM chip.

Furthermore, pipelined MAC operations are effectively enabled through temporary charge storage facilitated by the sampling capacitors.

The MAC operation within the FACIM architecture follows a precise four-steps: 1) The 9T1C SRAM-based CIM cell performs the initial partial MAC operations. 2) The ASA subsequently combines the 7-bit magnitude partial results and samples the corresponding signed and 7-bit magnitude components of the analog feature. 3) A zero-crossing-based [17], [18], [19] transmitter (ZCBT) then merges and amplifies these signed and 7-bit magnitude components, transmitting the resulting output voltage to the subsequent layer. 4) This transmitted output triggers charge redistribution in the subsequent layer, which efficiently initiates the next cycle of partial MAC operations in a pipelined manner.

Different from previous approaches [10], [11], [12], [13], [14], the proposed FACIM architecture eliminates quantization noise by enabling end-to-end analog signal propagation between neural layers. Targeting edge-AI device applications, the design effectively preserves feature integrity and significantly enhances energy efficiency by minimizing redundant data movement.

To validate the FACIM concept, a prototype macro implementing a two-layer fully connected ($784 \times 100 \times 10$) neural network was fabricated. As shown in Fig. 4(a), the prototype incorporates an input buffer and a decision circuit (comparator

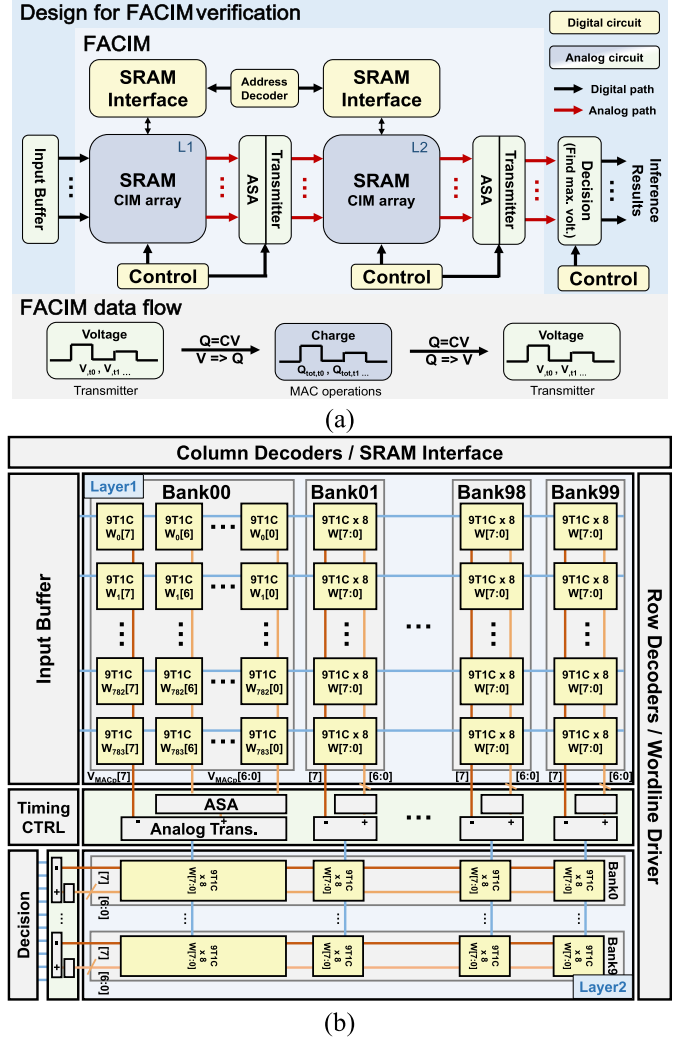


Fig. 4. (a) Structure and data flow of proposed FACIM macro. (b) Floorplan of the realized FACIM chip.

tree) to directly recognizes the binary MNIST dataset. This demonstration successfully confirmed the feasibility of analog signal transmission for triggering subsequent layers, thereby paving the way for fully analog neural network implementations. Fig. 4(b) shows the floorplan of the realized FACIM chip.

B. Proposed 9T1C SRAM-Based MAC Cell

Fig. 5 presents the schematic, layout, and operand table of the proposed 9T1C SRAM-based MAC cell. This cell executes partial MAC operations between stored weights and analog input features in the charge domain. Each cell comprises a 6T SRAM bit-cell for weight storage, a MAC capacitor (C_{MAC}), a pair of transmission gates (TGP and TGN) that charge C_{MAC} when the stored weight is "1", and an NMOS switch (SW) designed to keep C_{MAC} uncharged if the weight is "0".

The C_{MAC} is implemented using a MOM capacitor with a unit capacitance of approximately 1.5 fF. It stores the accumulated charge and couples it to the MAC line (MACL) for subsequent processing. To maximize integration density, the MOM capacitor adopts a compact three-layer sandwich

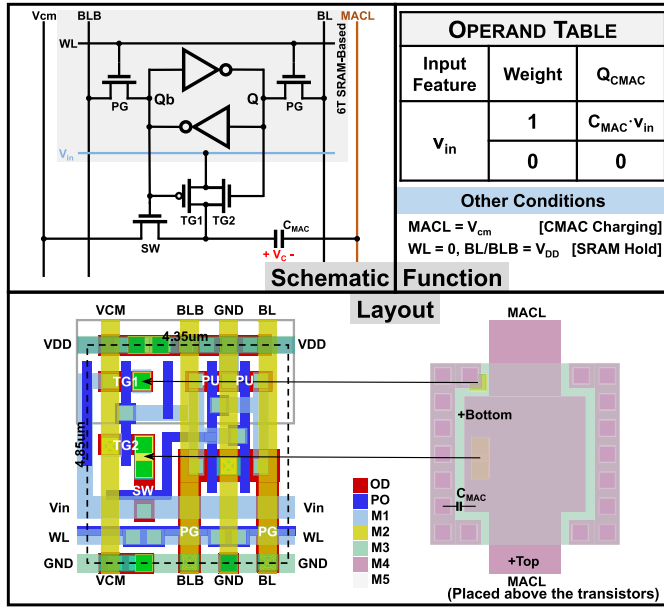


Fig. 5. Proposed 9T1C SRAM-based CIM cell design.

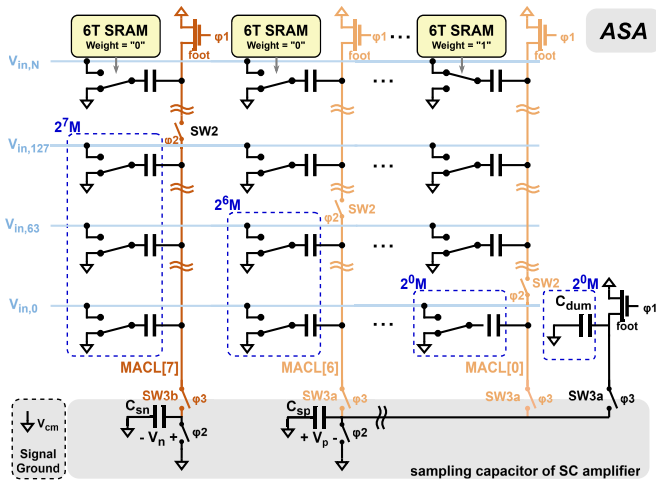


Fig. 6. Proposed analog shifts and adder design.

structure strategically placed above the MAC cell, thus incurring no additional area overhead. Furthermore, its bottom plate is designed to enclose the top plate, a feature that effectively reduces parasitic capacitance and significantly enhances signal integrity. This 9T1C SRAM-based MAC cell is fabricated using a 0.18- μm CMOS process following logic design rules and occupies a total area of 21 μm^2 (4.35 $\mu\text{m} \times 4.85 \mu\text{m}$). The additional peripheral circuitry for MAC operation accounts for approximately 40% of the total MAC cell area.

As shown in Fig. 7, the partial MAC operation in the proposed MAC cell is performed in two steps. In the first step, the MACL of each column is connected to $V_{cm} = 0.5 \cdot V_{DD}$ via the footer transistor. During this phase, analog input features (rail-to-rail) from the previous layer are stored in the C_{MAC} of cells where the stored weight is “1,” while the other C_{MAC} are kept uncharged (V_{cm}). In the second step, the footer is turned off, and the V_{in} is driven to V_{cm} . This action allows the redistribution of the charge stored in C_{MAC} .

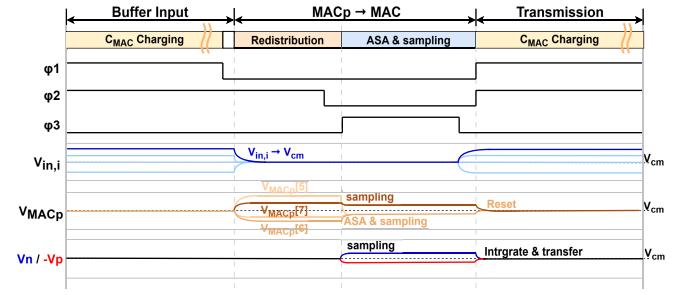


Fig. 7. Timing diagram for MAC operation in FACIM.

This redistribution creates a partial MAC result between the analog input features and the 1-bit weights. The voltage on k^{th} MACL ($V_{MACp}[k]$) resulting from this operation can be mathematically expressed as:

$$V_{MACp}[k] = \frac{-\sum_{i=1}^N (w_i[k] \cdot C_{MAC} \cdot V_{in,i})}{N \cdot C_{MAC} + C_p} \quad (1)$$

where N denotes the number of elements in the input feature vector, $V_{in,i}$ represents the rail-to-rail input feature at the i^{th} row, $w_i[k]$ signifies the k^{th} bit of the weight stored in the i^{th} MAC cell, C_{MAC} is the unit capacitor in each MAC cell, and C_p is the total parasitic capacitance on the MACL.

C. Proposed Analog Shifts and Adder

Fig. 6 illustrates the schematic of the proposed Analog Shifter-Adder (ASA). After the 9T1C MAC array completes partial MAC operations between 1-bit weights and analog input features, the resulting charges are uniformly stored in the C_{MAC} array within the charge domain. The ASA then partitions this C_{MAC} array based on binary weight scaling, ensuring that the charge stored in each capacitor column follows a 2^n ratio. When switch SW3 is turned on, charge sharing occurs to perform the 1-bit signed and 7-bit magnitude MAC operation. Both positive (V_p) and negative (V_n) components are subsequently captured by the sampling capacitor (C_{sp} / C_{sn}), as expressed by the following equations:

$$V_n = \frac{2^7 \cdot M \cdot C_{MAC} \cdot V_{MACp}[7]}{2^7 \cdot M \cdot C_{MAC} + C_s} \quad (2)$$

$$V_p = \frac{-\sum_{k=0}^6 2^k \cdot M \cdot C_{MAC,unit} \cdot V_{MACp}[k] - C_{dum} \cdot 0}{2^7 \cdot M \cdot C_{MAC} + C_s} \quad (3)$$

where $C_{dum} = C_{MAC}$ represents a dummy CIM cell employed to compensate for the number of CIM cells on the positive component. M is the quotient of N divided by 2^7 , and $[k]$ denotes the k^{th} bit weight. To minimize signal attenuation and reduce noise interference, 128M is chosen as the maximum integer value among the C_{MAC} array count. V_n represents the sign-bit MAC component, and V_p denotes the magnitude MAC component. Considering both sampling noise and additional noise sources, C_s is ultimately set to 150 fF.

As shown in Fig. 7, the MAC operation follows a pipelined sequence in the FACIM architecture. First, each MAC cell performs a partial MAC operation, and the resulting charge is

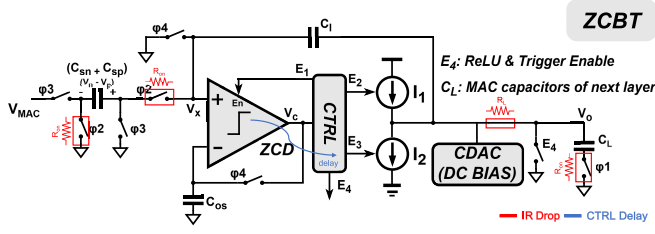


Fig. 8. Proposed zero-crossing based transmission design.

uniformly stored in the C_{MAC} array within the charge domain. Next, the ASA processes these stored charges to complete the 1-bit signed and 7-bit magnitude MAC operation. The resulting voltages are then stored in C_{sn} and C_{sp} , which are subsequently transmitted to the next neural layer via the ZCBT. Concurrently, the CIM cells begin the next cycle of partial MAC operations, thereby enabling continuous and efficient pipelined MAC operation in the analog dataflow.

D. Proposed Zero-Crossing Based Transmitter

Traditional analog transmission circuits typically rely on operational amplifiers (opamps), which suffer from high static power consumption and limited operating speed. As a result, in opamp-based FACIM systems, analog data transmission accounts for over 90% of the total power consumption. To significantly improve energy efficiency, this work adopts a Zero-crossing-based (ZCB) amplifier technique [17], [18], [19]. This method retains the closed-loop operation concept of conventional amplifiers but replaces the opamp with a current source, a Zero-crossing detector (ZCD), and control logic. As shown in Fig. 8, when the ZCD detects that the node voltage V_x crosses the virtual ground level V_{cm} , the analog signal transmission is completed, and the ZCBT is immediately shut down to prevent static power consumption.

The ZCD functions as a continuous-time comparator, making it well-suited for implementing the ReLU activation function in neural networks. By leveraging the ZCD, the system can detect whether the analog feature signal exceeds a predefined threshold before transmission. If the threshold is met, the analog feature is propagated to the next layer; otherwise, the ZCBT remains disabled. This mechanism significantly reduces static power consumption, drawing only minimal energy from the biasing circuitry. Compared to opamp-based FACIM designs, the proposed ZCB FACIM achieves a higher throughput and an improved energy efficiency.

As shown in Fig. 8, the ZCBT adopts a switched-capacitor (SC) amplifier architecture. Since the V_{MAC} signal typically exhibits a small amplitude, amplification is required to maintain a sufficient signal range. The amplification gain is determined by variations of the output feature. During the amplification phase, a current source charges the load capacitor (C_L), and the resulting voltage is coupled to the internal node V_x through a feedback capacitor (C_1). Once the ZCD detects that V_x has reached the virtual ground voltage, the current source is immediately turned off, thereby completing the zero-crossing event and halting further charging.

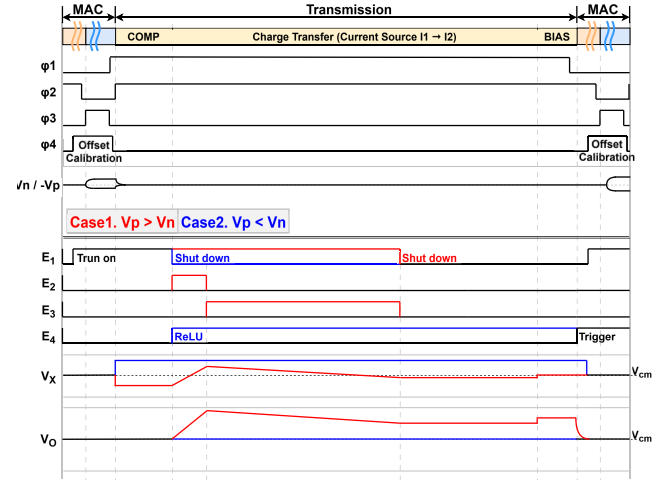


Fig. 9. Timing diagram for transmission in FACIM.

However, the ZCD cannot instantly shut off the current source at the exact moment when V_x reaches the virtual ground voltage, due to inherent detection delay (t_d). This results in a voltage overshoot (V_{over}), as described in Equation (4). Furthermore, the on-resistance of the switching transistors introduces additional voltage drop errors (V_e) during charge transfer, which can be quantified by Equation (5).

$$V_{over} = \frac{I}{C_x + C_L} \times t_d \quad (4)$$

$$V_e = \frac{C_{sp} + C_{sn}}{C_1} \times V_{IR2} + V_{IRL} + V_{IR1} \quad (5)$$

where I represents the current used to charge C_L , C_x is series combination of C_1 and $(C_{sp} + C_{sn})$, C_L corresponds to the C_{MAC} array in the next neural network layer, t_d denotes the delay caused by threshold detection, V_{IR1} and V_{IR2} represent the IR drop across the SW1 and SW2, and V_{IRL} represents the IR drop caused by the resistance of the MACL.

As shown in Equations (4) and (5), both the voltage overshoot (V_{over}) and the error voltage (V_e) are closely related to the charging current. Therefore, the ZCBT employs a coarse-fine dual current source scheme to ensure both operational speed and precision, as proposed in [17]. Additionally, the integrated CDAC can be used to cancel residual overshoot. Once the output feature transmission is complete, the controller triggers the charge redistribution of the next layer through control signal E_4 , initiating the next layer's partial MAC operation.

As illustrated in Fig. 9, the feature transmission process in the ZCBT is divided into four sequential phases. First, during the initialization phase, the ZCBT is reactivated, as it automatically powers down after each transmission to conserve static power. In this phase, an Auto-zero calibration [20] is performed, wherein the offset of the ZCD is stored in a capacitor (C_{os}), enabling automatic cancellation of ZCD's offset during the subsequent transmission.

In the comparison phase, the negative terminal of C_s is connected to the V_{cm} , coupling V_{MAC} to node V_x . The ZCD then determines whether V_x is negative, functioning as

an analog implementation of the ReLU activation function. If V_x is positive, the input signal is below the activation threshold, and the ZCBT remains shut down, keeping the output voltage (V_o) clamped to V_{cm} . If V_x is negative, indicating a valid feature, the ZCBT is enabled for transmission.

During the transmission phase, a coarse current source (I_1) rapidly charges the C_L . Upon detection of the zero-crossing event, a fine current source (I_2) is activated to suppress voltage overshoot. A CDAC subsequently injects DC voltage to cancel residual voltage error. Simulation results indicate that, even under PVT variations and device mismatch, the input reference voltage error remains within 500 μ V.

Finally, in the trigger phase, once feature transmission is complete, a control signal is generated to initiate charge redistribution in the next neural layer, thereby enabling the next-stage partial MAC operation in the analog pipeline.

It is important to note that, to ensure stable operation of the fine current source (I_2), it is enabled concurrently with the coarse current source (I_1) at the beginning of the transmission phase. Although this approach incurs additional energy consumption, it is crucial for maintaining the stability of the fine current source and thereby preserving the precision of the analog signal transmission.

IV. TRAINING-HARDWARE CONSISTENCY IN NEURAL NETWORK SYSTEMS

Analog CIM architectures offer notable advantages in speed and energy efficiency compared to their digital counterparts. However, their inference accuracy is often limited by non-ideal factors such as circuit mismatch and noise, which need to be carefully considered during system design.

The proposed FACIM architecture avoids quantization noises and ADC mismatches by performing feature extraction directly in the analog domain. Despite this, its performance is still influenced by several analog non-idealities, including C_{MAC} unit mismatch, sampling noise, and ZCBT-related PVT variations, mismatches, nonlinearity, and circuit noise.

To improve inference accuracy and mitigate the impact of these hardware-induced imperfections, this section analyzes these key non-ideal factors. It also details how hardware behavioral models are integrated directly into the neural network training process to evaluate the influence of these hardware non-idealities. This co-design approach ensures implementation, thereby enhancing the robustness and consistency between software training and hardware reliability of analog neural network systems.

A. CMAC Unit Mismatch

In charge-based CIM architectures, 1-bit MAC operations are typically implemented by controlling whether a capacitor is charged, thereby representing weight values of 1 or 0. However, different capacitor implementations exhibit varying degrees of sensitivity to PVT and device mismatch, which directly degrade the accuracy of partial MAC operations. To mitigate such sensitivity, this work adopts a capacitor-ratio-based MAC operation and employs MOM capacitors for the CMAC implementation, thereby enhancing robustness against PVT-induced variations.

Although MOM capacitors demonstrate superior PVT robustness, the mismatch between individual capacitors remains a critical issue. The matching characteristics of MOM capacitors depend strongly on the layout configuration and capacitor size. According to measurements reported in [21], the variation of 1.5 fF MOM capacitors fabricated in a 0.18- μ m process is typically within 0.3%, with the best matching achieved using a sandwich structure. However, as the technology node scales down, the capacitance density of MOM capacitors increases, but their relative variation can also escalate. For instance, [22] reports that in IBM's 32-nm CMOS SOI process, the variation of 1.2 fF MOM capacitors can reach approximately 0.8%. Therefore, the capacitor sizing must be carefully evaluated based on the specific process node during the design phase.

In the partial MAC operation, the mismatch (σ) contributed by each CMAC unit accumulates at the MACL. To maintain 10-bit output precision, the accumulated mismatch error must be less than $1/2^{(10+1)}$. Notably, capacitor mismatch constitutes a statistically independent and random error source. Consequently, the total variation accumulated from N mismatches grows only by $\sqrt{N}$. As a result, the final mismatch-induced V_{MACp} error scales as $(\sqrt{N}/N)\sigma$, leading to a mismatch requirement for each CMAC unit of approximately 0.05 $\sqrt{N}$ % (where N is the number of accumulated units).

In the ASA circuit, the mismatch from all CMAC units also contributes to the final MAC result. Since the ASA is constructed from 256M CMAC units, its total mismatch scales as $\sigma/(16\sqrt{M})$. To satisfy the 10-bit inter-layer analog transmission requirement, the total accumulated mismatch must remain below $1/2^{(10+1)}$. This constraint implies that the mismatch of each CMAC unit must be less than approximately 0.78 $\sqrt{M}$ % (where M is the quotient of N divided by 2^7).

Furthermore, noise is typically introduced during standard neural network training to mitigate the risk of overfitting. A judiciously designed noise level also helps increase the tolerance to computation losses caused by CIM hardware non-idealities. Therefore, we modeled the non-idealities of the actual hardware and integrated them into the neural network's co-training process. Specifically, as illustrated in Fig. 10, a 0.5% capacitor variation is treated as quantization error and injected into the model through a Gaussian noise layer to simulate and verify the impact of MOM capacitor mismatch. This retraining approach aligns the training environment more closely with the actual hardware behavior, thereby enhancing the system's overall robustness.

B. Sampling Noise

Although neural networks can tolerate some feature noise, excessive noise still results in a significant degradation of model accuracy. Therefore, effectively suppressing noise generated during analog MAC operations is crucial for maintaining overall system accuracy.

In charge-domain analog MAC operations, sampling noise is one of the primary non-idealities. During sampling, thermal (kT/C) noise is stored on the capacitor, where noise amplitude is inversely proportional to capacitance. If the sampling capacitor is too small, the resulting noise voltage significantly

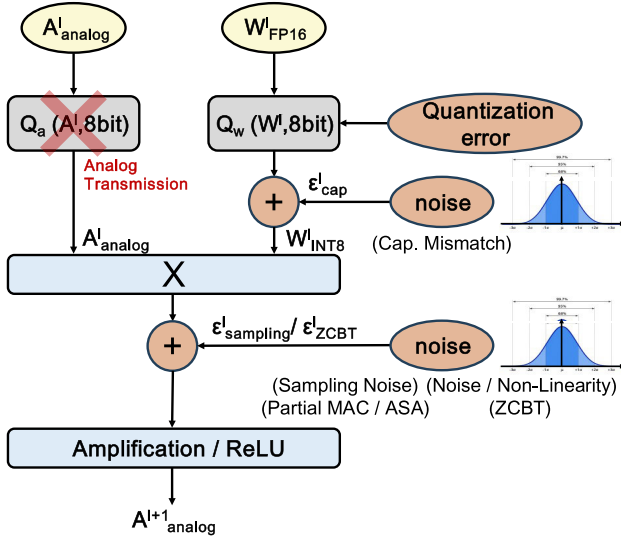


Fig. 10. Neural network model w/ analog behavior.

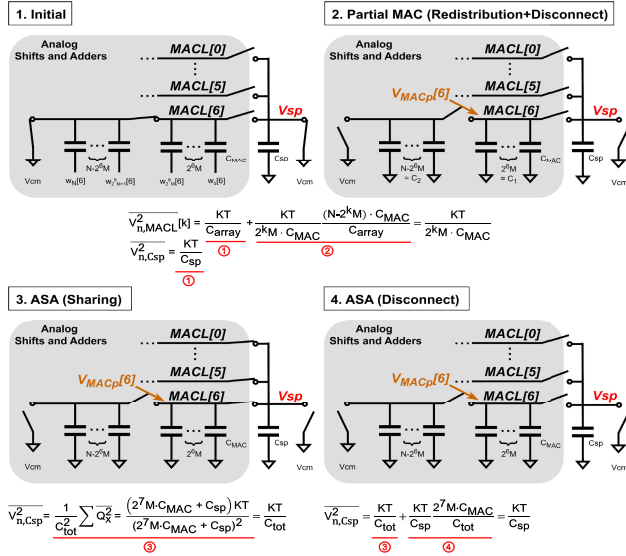


Fig. 11. Sampling noise analysis of analog MAC operation.

increases, degrading signal quality and causing a noticeable drop in neural network inference accuracy.

To address this, this section models and analyzes the sampling noise generated during charge-domain analog MAC operations. Based on the required accuracy and power consumption, an appropriate sampling capacitor size is chosen to achieve an optimal trade-off. Furthermore, as illustrated in Fig. 10, the noise model is integrated into the neural network training by injecting equivalent sampling noise. This approach aligns the training conditions more closely with the actual hardware behavior.

As shown in Fig. 11, the MAC operation in the FACIM architecture can be divided into four steps for noise analysis.

First, before performing the partial MAC operation, the C_{MAC} array and the sampling capacitor (C_s) must be initialized through the footer switch. However, during this initialization phase, thermal noise from the footer introduces additional

sampling noise into both the C_{MAC} array and C_s , as shown below:

$$\overline{V_{n,initial}^2} = \frac{KT}{C_x} \quad (6)$$

where K is Boltzmann's constant, T is the absolute temperature, C_x denotes either C_{array} or C_s , and C_{array} signifies the total capacitance on the MACL.

Second, before the ASA performs charge sharing, the MAC capacitor array must be disconnected according to the weight ratios. During this disconnection process, additional noise is introduced, as reported in [23] and [24] and expressed as follows:

$$\overline{V_n^2} = \frac{KT}{C_1} \frac{C_2}{C_1 + C_2} \quad (7)$$

where $C_1 + C_2$ represents the total capacitance on the MACL, C_1 denotes the portion of the capacitor array disconnected for ASA based on weight ratios, and C_2 refers to the remaining connected capacitance.

Third, after the MAC capacitor array is disconnected according to the weight ratios, the ASA performs charge sharing to accumulate the positive and negative charge components, corresponding to the positive and negative MAC results. These are then sampled by two dedicated sampling capacitors, C_{sp} and C_{sn} , respectively. During the charge-sharing process, noise charges are also transferred to the output node. Since these noise charges are uncorrelated, the variance of the total shared noise charge is given by:

$$\overline{V_{n,shared}^2} = \frac{1}{C_{tot}^2} \sum Q_x^2 = \frac{1}{C_{tot}^2} \sum C_x^2 V_x^2 \quad (8)$$

where C_{tot} is the total capacitance involved in the charge sharing process, and Q_x^2 represents the variance of the noise charge contributed by each capacitor.

Finally, after disconnecting from the MACL, the sampling capacitors C_{sp} and C_{sn} perform individual sampling operations, each introducing sampling noise as described in Equation (7). According to the procedure illustrated in Fig. 11, whether the MAC operation result corresponds to the positive or negative component, the final output is inevitably affected by thermal (KT/C_s) noise. This noise directly impacts the accuracy of the accumulated result and may further accumulate in subsequent layers. Therefore, careful consideration of the sampling capacitor size and noise impact is essential during system design and noise modeling to ensure the overall reliability of the analog MAC operation. Based on this consideration, the proposed FACIM adopts 150-fF sampling capacitors (C_{sp} & C_{sn}), resulting in a calculated noise of approximately 164 μ V, which corresponds to about one-third of the overall transmission noise budget and is included in the ZCBT noise simulation.

C. Noise of the ZCBT

In addition to the noise generated during MAC operations, the noise sources and linearity of the analog signal transmission circuits are also crucial for the overall performance of the FACIM system. Since this architecture implements MAC operations in the charge domain, it's particularly susceptible to coupling noise. Without proper suppression, such noise can significantly degrade transmission accuracy.

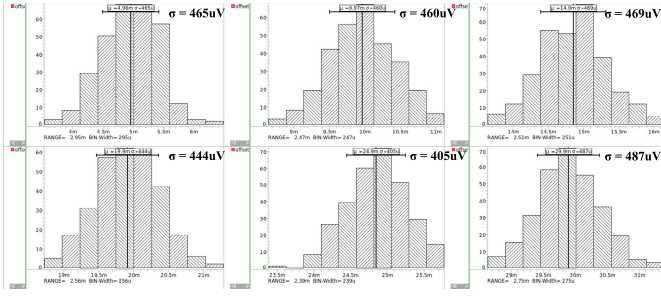


Fig. 12. ZCBT and Sampling noise simulations (post-layout).

To address this, the proposed ZCD adopts a current-mirror OTA to implement a continuous-time comparator. This design effectively suppresses kickback noise generated during ZCD switching, improving input feature reliability.

Despite these measures, the ZCBT still faces multiple noise sources during operation, including the thermal noise from the ZCD circuit, the kT/C noise introduced during the integration capacitor's reset phase, current-induced noise from the integration current source I_2 , and timing jitter caused by digital control logic transitions. The variance of the ZCBT's integrated noise voltage is referred to the input, as shown below:

$$\overline{V_{n,ZCBT}^2} = \overline{V_{n,ZCD}^2} + \frac{1}{A_{SC}^2} (\overline{V_{n,Cint}^2} + \overline{V_{n,I2}^2} + \overline{V_{n,ctrl}^2}) \quad (9)$$

where $V_{n,ZCD}$ denotes the noise voltage of the ZCD, A_{SC} represents the gain of the ZCBT, $V_{n,Cint}$ refers to the sampling noise of the integration capacitor, $V_{n,I2}$ corresponds to the noise voltage induced by the integration current I_2 , and $V_{n,ctrl}$ accounts for the jitter introduced by the current source disable. This equation indicates that the ZCD is the dominant noise source, as other contributions are attenuated by the gain of the ZCBT. As shown in Fig. 12, post-layout simulations were conducted to evaluate the impact of ZCBT-related noise, and the standard deviation (σ) of the input-referred noise voltage was found to be less than 500 μV .

D. Mismatch Effect of the ZCBT

In addition to the significant impact of noise on ZCBT performance, inter-channel circuit mismatch is also a critical factor affecting overall system accuracy. While the mismatch in the ZCD can be treated as an input offset and effectively suppressed using auto-zero techniques, mismatches from controller delay and current sources are more difficult to eliminate and often require off-chip calibration algorithms for compensation.

Similarly, in neural network applications, mismatch characteristics can be incorporated into model training to improve system adaptability and robustness. However, this approach significantly increases training cost and may hinder network convergence. Therefore, it is essential to carefully evaluate and mitigate the impact of mismatches during the circuit design process to ensure stable and reliable system performance.

It is worth noting that the errors caused by controller delay and current source mismatch behave similarly to noise and are

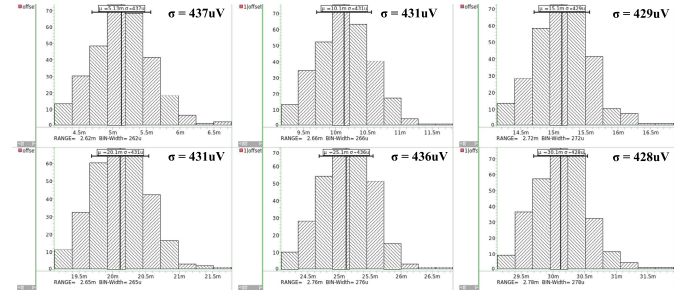


Fig. 13. ZCBT mismatch simulations (post-layout).

effectively suppressed by the gain of the ZCBT. As a result, their impact on system accuracy is relatively smaller compared to the mismatch in the ZCD. As shown in Fig. 13, a Monte Carlo simulation was conducted to evaluate the mismatch in the ZCBT, and the resulting input-referred mismatch voltage standard deviation (σ) was found to be less than 500 μV , confirming the robustness of the ZCBT.

E. Linearity Analysis of the ZCBT

Finally, the linearity performance of the ZCBT under different PVT conditions is considered in detail. The ZCBT circuit used in this paper can achieve nearly rail-to-rail signal transmission when forwarding positive features. This is due to its transmission process being divided into a coarse phase and a fine phase for improving precision. Even though the p-type current source (I_1) in the coarse phase experiences a small V_{ds} , causing variations in the charging current, the n-type current source (I_2) in the fine phase operates over a wide V_{ds} range, ensuring that I_2 remains very stable. As shown in Fig. 14, post-layout simulations under PVT variations indicate that the maximum linearity error of the ZCBT, referred to the input, remains below 100 μV .

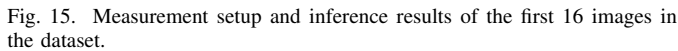
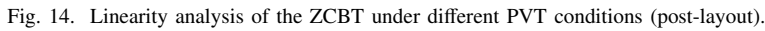
V. PERFORMANCE AND MEASUREMENT RESULTS

A. Chip Implementation Overview

The FACIM chip is fabricated as an application-specific integrated circuit (ASIC) designed for edge-AI pre-processing, with its core concept focusing on feature extraction in the analog domain to reduce unnecessary data transmission. The chip integrates a two-layer analog neural network (FC: $784 \times 100 \times 10$), an input buffer, and an analog decision circuit (comparator tree), and is implemented using TSMC 0.18- μm CMOS technology with a standard 1P6M process. The entire chip is fully custom-designed following logic design rules.

B. FPGA-Based Measurement and Verification Method

Since integrating 784x DACs or a CMOS image sensor into the chip is constrained by area limitations, an FPGA is adopted as the front-end data interface to deliver the binary MNIST dataset to the fabricated chip for validation, as shown in Fig. 15. In conjunction with the two-layer neural network implemented on the chip, this configuration is utilized to verify

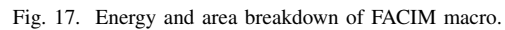
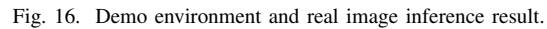


charge-domain MAC operations and the feasibility of analog inter-layer signal transmission. These measurements also indirectly demonstrate that the proposed FACIM architecture can accept analog feature inputs and perform analog-domain feature extraction across neural network layers.

Fig. 15 also presents the actual recognition results of the first 16 test images from the binary MNIST dataset processed by the FACIM, offering an intuitive demonstration of the inference correctness enabled by the proposed analog inter-layer transmission mechanism.

By inputting all 10,000 test images and calculating the final classification accuracy, the chip performance can be back-inferred at the system level. This methodology enables precise estimation of the accumulated error range of analog signals in critical analog operations, including charge-domain MAC, signal amplification, and inter-layer transmission. Based on this inferred accuracy, the overall circuit performance of the proposed FACIM can be further evaluated.

Furthermore, as shown in Fig. 16, an FPGA combined with a camera platform is employed to acquire real-time image data. The captured images are preprocessed at the front end



and converted into the binary MNIST format before being fed into the chip for inference. This experiment successfully validates the fabricated chip using real-world image data and further demonstrates the feasibility of the proposed FACIM architecture. Fig. 16 also presents the recognition results of the FACIM for real-world images.

C. Overall Chip Performance

Fig. 17 presents a detailed breakdown of the FACIM chip’s energy consumption and area distribution. Overall energy consumption is dominated by the signal transmission blocks, with the ZCBT accounting for 66% and the input buffer contributing 27%, while the MAC array consumes only

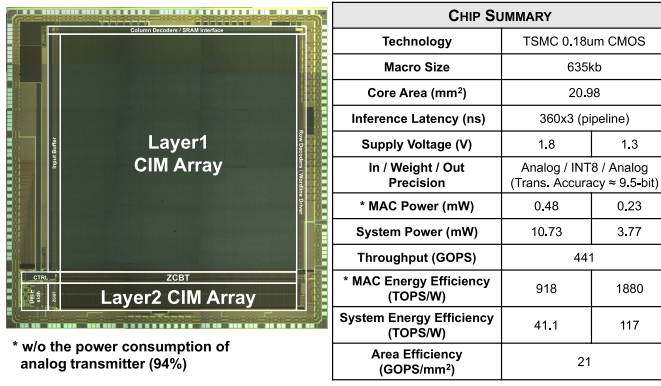


Fig. 18. Die photograph and chip performance summary.

approximately 6% of the total energy. Regarding area distribution, the MAC array occupies the largest portion of the die area, accounting for 86%, while the SRAM interface and input buffer occupy 5% and 4%, respectively. Within the MAC array, the SRAM itself utilizes 60% of the area, with the remaining 40% dedicated to the MAC circuitry.

Fig. 18 shows the die photograph and summarizes the performance of the FACIM chip. The macro occupies 20.98 mm² and provides a storage capacity of 635 kb. Operating at a standard supply voltage of 1.8 V, the chip achieves an inference latency of 1080 ns per image (360 ns per layer across three layers), with an energy efficiency of 918 TOPS/W for MAC operations and 41.1 TOPS/W for full-network inference, and a throughput of 441 GOPS.

To further reduce power consumption, the supply voltage can be lowered to 1.3 V without compromising operating speed. Under this condition, the energy efficiency improves to 1880 TOPS/W for MAC operations and 117 TOPS/W for full-network inference. The normalized area efficiency reaches 97.76 GOPS/mm². When evaluated on binary MNIST classification, the chip achieves an accuracy of 93.5%.

D. Accuracy and Noise Analysis

Full-precision models on the binary MNIST dataset achieve around 97% recognition accuracy, whereas incorporating 8-bit weight quantization and 10-bit transmission noise reduces the software baseline accuracy to about 94%. Furthermore, to analyze hardware nonidealities, behavioral-level models are constructed for both the ADC-based CIM and the proposed FACIM. As shown in Fig. 19, the effects of the ADC effective number of bits (ENOB) and the FACIM transmission noise on recognition accuracy is evaluated through controlled simulations.

An important observation can be drawn from the simulation results: since quantization noise exhibits strong data dependence, whereas the transmission noise in FACIM is closer to a normal distribution, the proposed FACIM achieves higher recognition accuracy than the conventional ADC-based CIM under comparable noise magnitudes. Moreover, the simulations verify that the quantization noise associated with the MSB dominates overall CIM system performance, indicating that a higher output bit count is not the sole determinant of enhanced performance; however, this behavioral analysis does

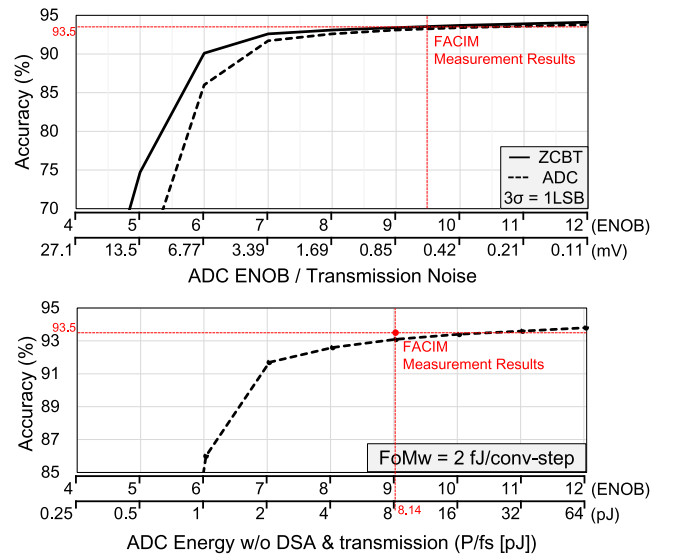


Fig. 19. ADC ENOB and transmission noise versus inferred accuracy.

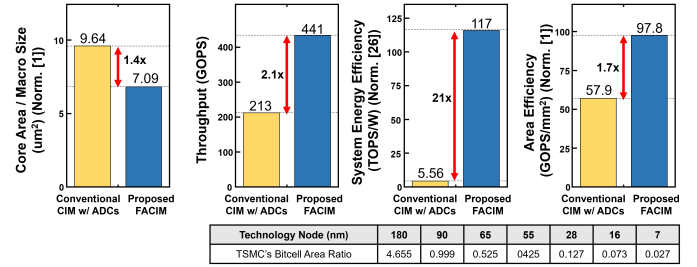


Fig. 20. Performance comparison between FACIM and ADC-based charge-domain CIM [11].

not yet account for the effect of channel mismatch among the ADCs.

The measured recognition accuracy of the proposed FACIM is 93.5%, which is slightly lower than that of the ideal 8-bit quantized model (94%). By back-inference, the equivalent transmission precision is estimated to lie between 9 and 10 bits, and this discrepancy is likely due to the accumulation of subtle circuit nonidealities.

In terms of energy efficiency, the proposed FACIM consumes approximately 8.14 pJ per operation. Even if the ADC in a conventional ADC-based CIM is assumed to achieve an ultimate Walden figure of merit (FoM) of 2 fJ/conv-step, the corresponding ENOB would still be limited to about 9 bits. In addition, this energy consumption does not include the extra overhead of the DSA and digital data transmission. These comparisons indicate that the proposed FACIM still exhibits a significant system-level energy-efficiency advantage.

E. Comparison With State-of-the-Art CIM Macros

Fig. 20 presents a performance comparison between the proposed FACIM design and an ADC-based charge-domain CIM [11]. By stacking the MOM capacitor directly above the MAC cell, the proposed FACIM incurs no additional area overhead. Furthermore, by adopting analog signal transfer instead of ADC-based data conversion, it significantly enhances energy efficiency. As a result, the proposed FACIM

TABLE I
COMPARISON TABLE OF THE PROPOSED FACIM MACROS WITH STATE-OF-THE-ART CIM MACROS

PREVIOUS WORK	JSSC 2023 [11]	JSSC 2024 [12]	JSSC 2024 [13]	S'VLSI 2022 [25]	This work
Technology	28nm	65nm	28nm	40nm	180nm
Memory Type	SRAM	SRAM	SRAM	RRAM	SRAM
SRAM Layout Rule	Aggressive Rule	Aggressive Rule	Aggressive Rule		Logic Rule
MAC Method	Charge sharing	Current discharge	Time domain	Current discharge	Charge coupling
Macro Size	384kb	64kb	1Mb	131kb	635kb
Core Area (mm ²)	0.47	1.71	1.22	0.62	20.98
Supply Voltage (V)	0.7	0.62 - 1 (Digital) 1 (Analog)	0.65	0.9	1.3
SA / ADC (bit)	5	4	7 (TDC)	N/A	N/A
In / Weight / Out Precision (bit)	8 / 8 / 20	8 / 8 / 20	8 / 8 / 22	Analog / 2 / Analog	Analog / INT8 / Analog (Trans. Accuracy $\approx$ 9.5-bit)
System Power (mW)	9.38	84.1	26.71	1.94	3.77
Throughput (GOPS)	213.4	100	1050	13.93	441
Energy Efficiency (TOPS/W)	22.75	2.75	39.31	26.97	117
Area Efficiency (GOPS/mm ²)	456	58.5	860	22.5	21
Normalized [26] Energy Efficiency (TOPS/W)	5.56	1.27	9.61	8.64	117
Normalized [1] Area Efficiency (GOPS/mm ²)	57.91	6.31	109.22	5.23	97.76

$$\text{Norm. EEf} = \text{EEf} * (1 + 30\%)^{\log_2 \left(\frac{\text{node}_{\text{node}}}{180\text{nm}} \right)^2} \text{ (TOPS/W)}$$

achieves a 1.4x reduction in normalized area overhead, a 2.1x increase in throughput, a 21x improvement in normalized system energy efficiency, and a 1.7x improvement in normalized area efficiency.

Table I compares the proposed FACIM macro with recent CIM works [11], [12], [13], [25]. Despite being implemented in different technology nodes, FACIM demonstrates superior energy efficiency compared to previous CIM designs supporting 8-bit MAC operations. Furthermore, normalized by TSMC's SRAM bit-cell area ratio [1], the normalized area efficiency of FACIM, achieved without relying on aggressive layout rules, outperforms most previous CIM designs supporting 8-bit MAC operations.

VI. CONCLUSION

This paper presents a Fully Analog Computing-in-Memory (FACIM) macro that directly extracts features in the analog domain, thereby avoiding feature degradation caused by quantization noise. The proposed design targets edge-AI pre-processing by reducing redundant data transmission and lowering both the number and energy consumption of ADCs.

The FACIM employs the proposed Analog Shifter-Adder (ASA) to support INT8 MAC operations and integrates a Zero-Crossing-Based Transmitter (ZCBT) with ReLU functionality, enabling high energy efficiency and throughput even at low supply voltages. Chip measurement results validate FACIM's superior performance for front-end feature extraction in small neural networks, achieving significant improvements in area efficiency, energy efficiency, and processing speed compared with conventional ADC-based CIM architectures.

Furthermore, the training-hardware consistency approach effectively mitigates the impact of process variation and analog circuit non-idealities on inference accuracy, ensuring system robustness.

Future work will focus on scaling FACIM to support larger and more complex neural network models by developing memory units capable of temporarily storing analog signals.

The proposed architecture also demonstrates strong potential for deployment in IoT sensing applications, offering a fully analog neural network hardware solution for edge-AI devices.

ACKNOWLEDGMENT

The authors would like to acknowledge chip fabrication support provided by Taiwan Semiconductor Research Institute (TSRI), Taiwan.

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